SOURCES - PROJECT

Documentation

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1. The voltage regulator

A voltage regulator is an electronic circuit that automatically maintains a constant output voltage, regardless of changes in input voltage or load conditions. It ensures that the output voltage of a power supply or battery remains within a specified range, which is crucial for the proper operation of electronic devices. The block diagram is present below (Figure 1), and further details are presented in this project.

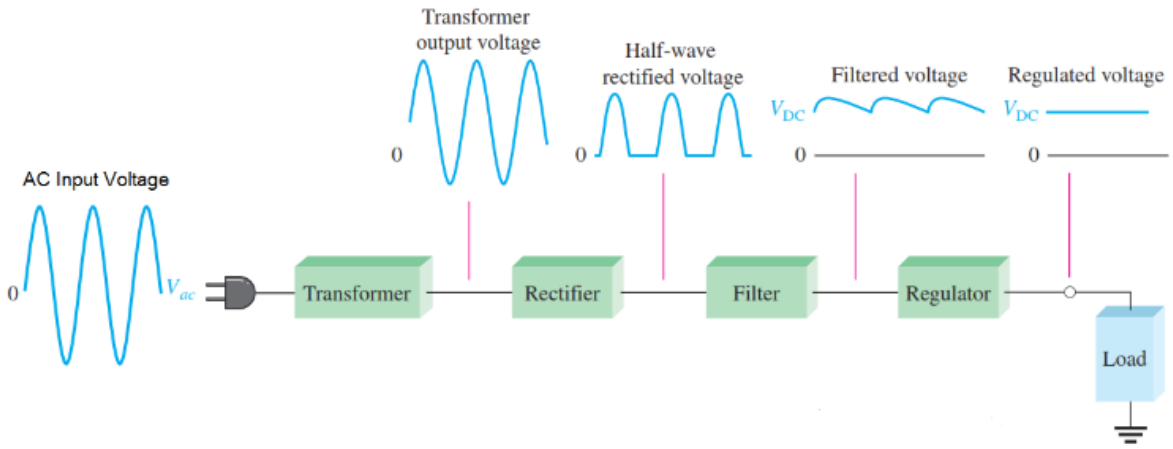


Figure 1 The block scheme

In summary, a voltage regulator is an essential component in electronic circuits, ensuring that devices receive a stable and constant power supply.

1.1. The transformer

A transformer is a device that transfers electric energy from one alternating-current circuit to one or more other circuits, either increasing (stepping up) or reducing (stepping down) the voltage. Transformers are employed for widely varying purposes; e.g., to reduce the voltage of conventional power circuits to operate low-voltage devices, such as doorbells and toy electric trains, and to raise the voltage from electric generators so that electric power can be transmitted over long distances. The role of the transformer (Figure X) is to reduce the value of the sinusoidal voltage from the power supply to the desired voltage. (Here, *Lprimary*, *Lsecondary* are the inductances of the two coils).

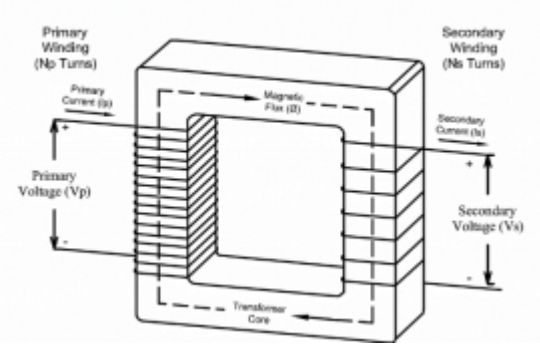


Figure 2. The transformer

1.2. The full-bridge rectifier

In our case, the voltage is sinusoidal, and since we want a DC value as the output, we need a full-bridge rectifier to get rid of the signal's negative components. This kind of single-phase rectifier (Figure 3) creates the desired output by connecting four separate rectifying diodes in a closed loop "bridge" configuration.

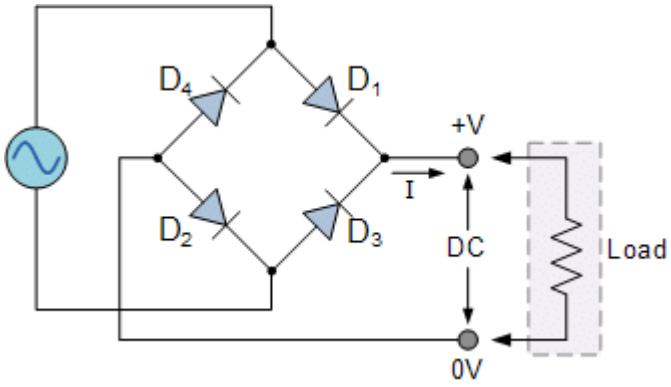


Figure 3 The full-bridge rectifier

The four diodes in Figure 4 designated D1 to D4 are placed in "series pairs" for the positive half-cycle, with only two diodes conducting current for each half-cycle. Diodes D1 and D2 are conducted in series, while diodes D3 and D4 are reverse biased and the current flows through the load as illustrated below during the positive half cycle of the supply.

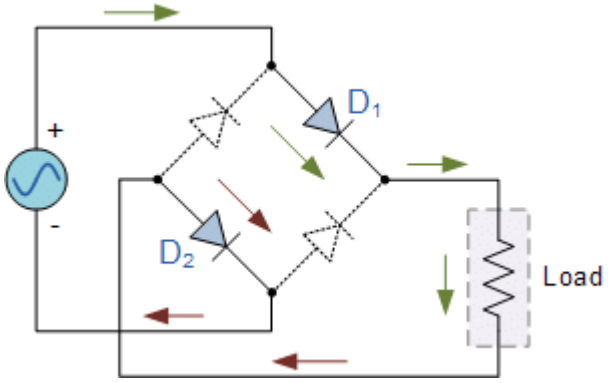


Figure 4 Negative half-cycle

Diodes D3 and D4 conduct in series during the negative half-cycle of the supply in Figure 5, while diodes D1 and D2 switch "OFF" because they are now reverse biased. The direction of the current flowing through the load is the same as previously.

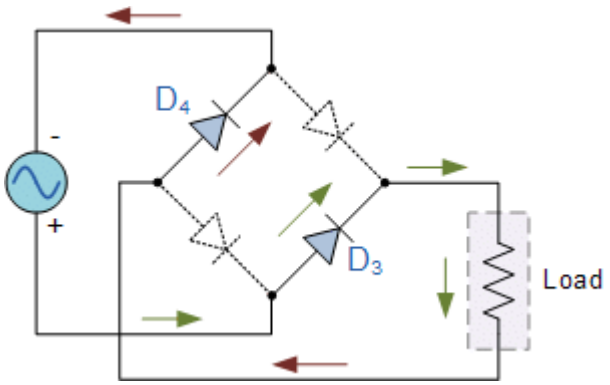


Figure 5.Positive Half-Cycle

The ideal waveform after the diode bridge will look like in Figure 6. In reality, twice the voltage drops across the diodes necessary for the current to pass will be lost during this rectification process.

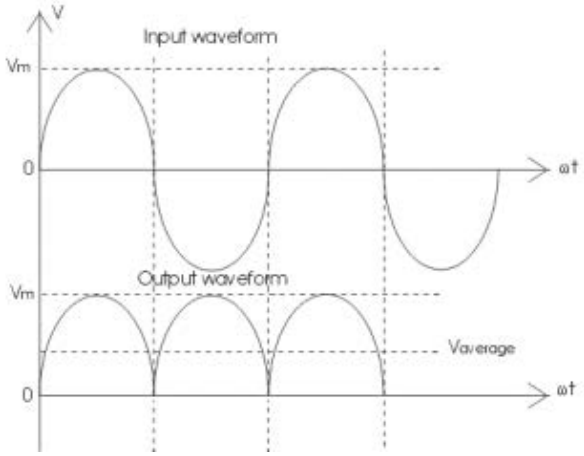


Figure 6.Input and Output signal

1.3. The capacitive filter

A capacitive filter, also known as a capacitor filter, is an electronic circuit that uses capacitors to filter out unwanted frequencies from a signal or power supply. It is an essential component in many electronic circuits, helping to ensure stable and reliable operation of electronic devices. Therefore, the capacitor uses its stored charge to deliver the necessary current. We can observe this phenomenon in the figure below (Figure X).

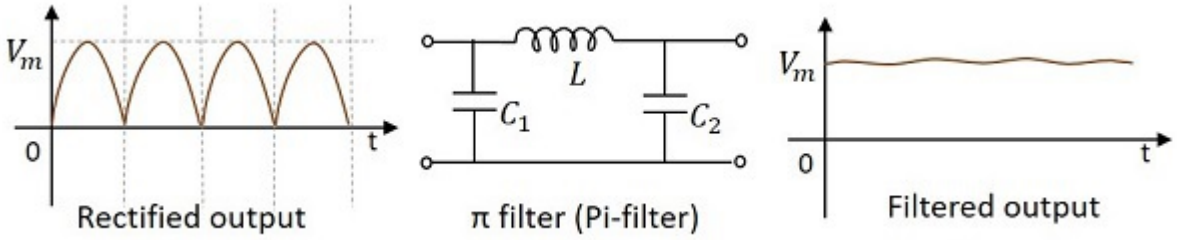


Figure 7. Filtered signal

2.1. The Linear Regulator

A linear regulator is a type of voltage regulator used in electronics that keeps the voltage constant. The regulator produces a constant voltage output by varying its resistance in response to changes in the load and the input voltage. .[1] In our project the regulator should have an output of 12V with a 500mA current and a efficiency of 60%.

2.2. The IC regulator

Three-terminal devices known as integrated circuit voltage regulators (IC voltage regulators) produce a steady DC output voltage regardless of the input voltage, output load current, or temperature. Furthermore, DC/DC converters control power buses and offer noise isolation. The output voltage of each kind of IC voltage regulator can be fixed or changed to a value inside a predetermined range.[2] In our project the IC should have a 5V output with 500mA current and a efficiency of 40%.

2.3. The buck converter

A DC-to-DC converter that increases current from its input (supply) to its output (load) while decreasing voltage is known as a buck converter or step-down converter. It belongs to the switched-mode power supply class. The buck converter, we use it to step down the output voltage from the full wave rectifier fto 3.3V with a current of 2A and a efficiency of 90%.

3. Circuit efficiency

Having the two values of power dissipation at the input and the output, we can calculate the circuit's efficiency using the following formula. Calculating efficiency is essential to understand how well the circuit utilizes the available energy and how much energy is lost as heat.

The efficiency of the circuit (η) can be determined by the ratio of the useful power at the output to the total power at the input, expressed as a percentage. The formula used is:

$$\eta = \frac{P_{output}}{P_{input}} * 100\%$$

where: P_{output} represents the average power dissipation at the circuit output and P_{input} represents the average power dissipation at the circuit input.

Also for the circuit initial calculations I made the following calculus:

1. $P_{active\ _secondary\ _1} = U_1 * I_1 = 12V * 0.5A = 6W$

$$\eta_1 = 60\% \Rightarrow \frac{P_{out\ _regulator\ _1}}{P_{out\ _rectifier\ _1}} = 60\% \Rightarrow P_{out\ _rectifier\ _1} = \frac{6}{0.6} = 10W$$

2. $P_{active\ _secondary\ _2} = U_2 * I_2 = 5V * 0.5A = 2.5W$

$$\eta_2 = 40\% \Rightarrow \frac{P_{out\ _regulator\ _2}}{P_{out\ _rectifier\ _2}} = 40\% \Rightarrow P_{out\ _rectifier\ _2} = \frac{2.5}{0.4} = 6.25W$$

3. $P_{active\ _secondary\ _3} = U_3 * I_3 = 3.3V * 2A = 6.6W$

$$\eta_3 = 90\% \Rightarrow \frac{P_{out\ _regulator\ _3}}{P_{out\ _rectifier\ _3}} = 90\% \Rightarrow P_{out\ _rectifier\ _3} = \frac{6.6}{0.9} = 7.(3)W$$

For the total power dissipation at the output we have the sum all the results above:

$$P_{out_total_rectifier} = P_{out_rectifier_1} + P_{out_rectifier_2} + P_{out_rectifier_3} = 10W + 6.25W + 7.33W = 23.58W$$

4. LTSpice implementation

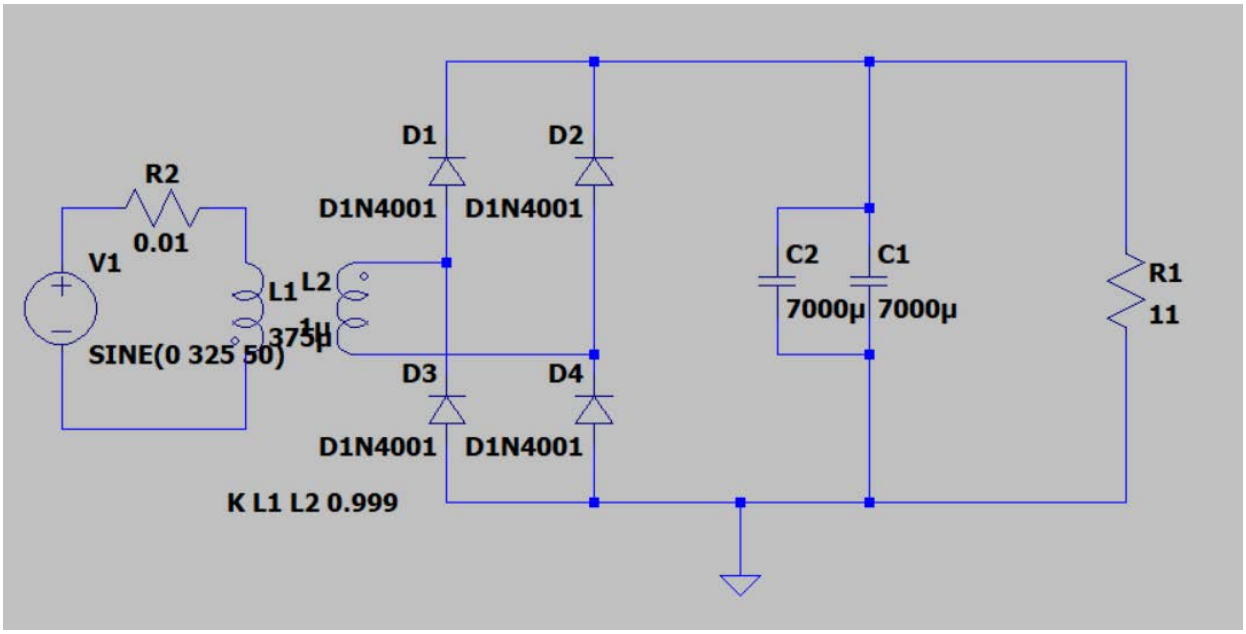


Figure 8 The LTSpice implementation

In this circuit, a 220V AC voltage source with a frequency of 50 Hz is used to reflect the European standard power supply specifications. This AC source is connected to a transformer, which is represented by the coupled inductors L1 and L2 with a mutual inductance for the program to consider them a transformer, allowing for voltage conversion suitable for the rectifier stage. The transformer ensures safe isolation and steps down the voltage to a level appropriate for rectification and filtering.

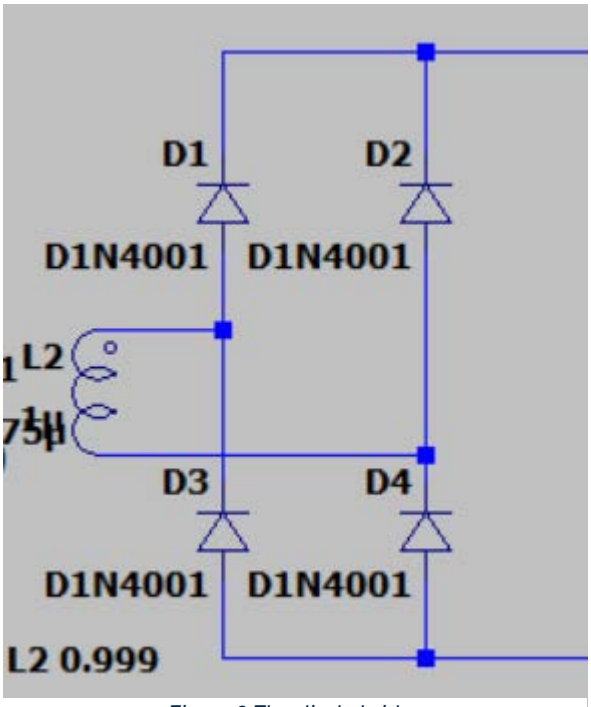


Figure 9 The diode bridge

For the rectification process, D1N4001 diodes have been used in a bridge configuration. These diodes were selected because they offer a forward current rating of 1A and a reverse current rating of 5uA, making them suitable for handling the current requirements of the circuit while providing reliable performance.

Ie used the 2 capacitors in prallel with the standard value of 7000 uF for the ripple voltage that comes out of the rectirfier to be as small as possible because it is a common value used for simulations.

$$C := \frac{\pi}{2 \cdot 0.031 \cdot 2 \cdot \pi \cdot 50 \cdot 10.92} = 0.0148$$

Figure 10 Filter calculus

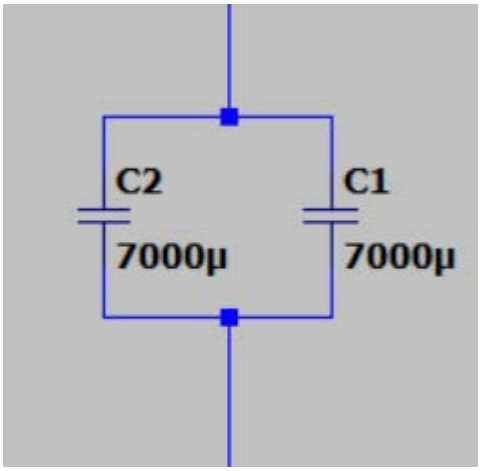


Figure 11 Filter implementation

5.Outputs

5.1. The linear regulator

The transformer's secondary coil should have a minimum voltage of 12V, but it's important to account for a 1.2V drop brought on by the transistor in the series regulator and an additional 1.4V drop brought on by the full-wave full-bridge rectifier's two diodes operating simultaneously. Therefore, I got the following calculus:

$$12V + 1.2V + 0.7 + 0.7 = 14.6V$$

We can consider this to be the voltage in the secondary coil and round it up to 15V.

Electronic equipment cannot operate properly without constant supply voltages. A rectifier with capacitive filtering has two components in its output voltage: a variable component and a DC component that is dependent on the network voltage. Furthermore, as the load current rises, this voltage significantly decreases and is affected by temperature changes.

The circuit schematic:

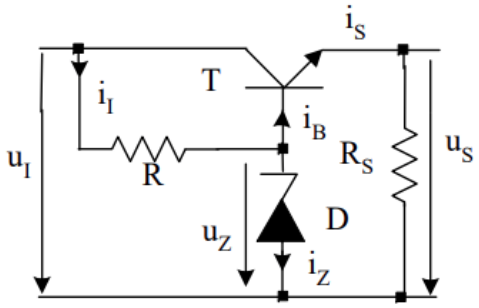


Figure 12 Linear regulator

5.1.1. The operating principle

The following is the series voltage regulator's working principle:

• Based on transistor T, the parametric regulator (R and D) produces a roughly constant voltage (uz).

• Transistor T acts as a comparator, comparing the full output voltage u_s to the reference voltage u_Z at the base: $u_{BE} = u_Z - u_S$. The error voltage that regulates the adjustment of the transistor's (the regulating element's) equivalent resistance value is obtained from this comparison.

• The output voltage and the voltage u_{BE} are out of phase. As a result, the transistor functions as an error amplifier when the output voltage rises and the current i_C through it falls. As a result, transistor T's voltage rises as well.

5.1.2. Designing the regulator

To obtain the wanted result I started by choosing a Zenner diode that will fulfill the condition of $U_Z = U_{BE} + U_S$. Therefore we need a diode with the capacity to provide 13V.

For the Ballast Resistance I used the formula:

$$R_{ballast} = \frac{U_{in_regulator_1} - U_z}{I_z + I_S(\beta_F + 1)}$$

Also to determine the current in the base of the transistor I used:

$$I_B = \frac{I_C}{\beta + 1} = \frac{0.5A}{100 + 1} = \frac{0.5}{101} = 0.00495A = 4.95mA$$

And for the voltage I used SMath to do the calculations:

$Uin_min := 16$

$Uz := 13$

$Uin_max := Uin_min + 0.579$

$Ib := 4.95\ mA$

$Iz := 5\ \ mA$

$Uballast_min := Uin_min - Uz = 3$

$Uballast_max := Uin_max - Uz = 3.579$

$Ir_ballast := Ib + Iz = 9.95\ mA$

$$Rballast_min := \frac{Uballast_min}{Ir_ballast} = 0.3015\ k\Omega$$

$$Rballast_max := \frac{Uballast_max}{Ir_ballast} = 0.3597\ k\Omega$$

Figure 13 Linear Regulator calculus

The final circuit looks like this:

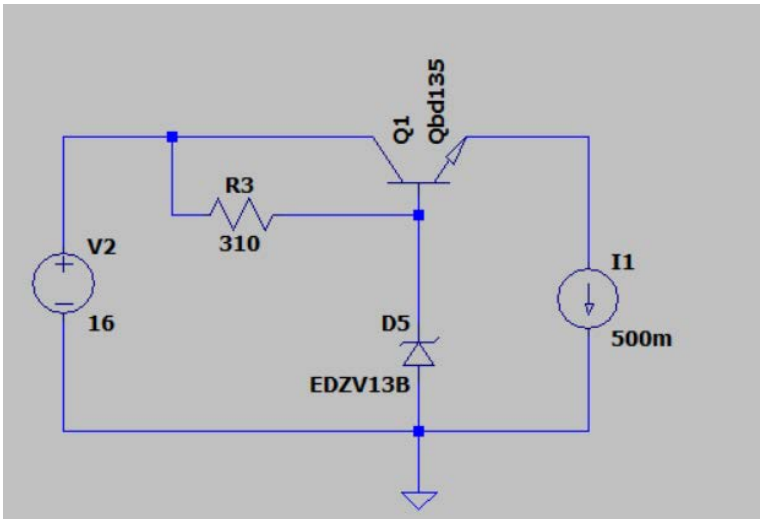


Figure 14 LTSpice implementation of the regulator

The simulations results:

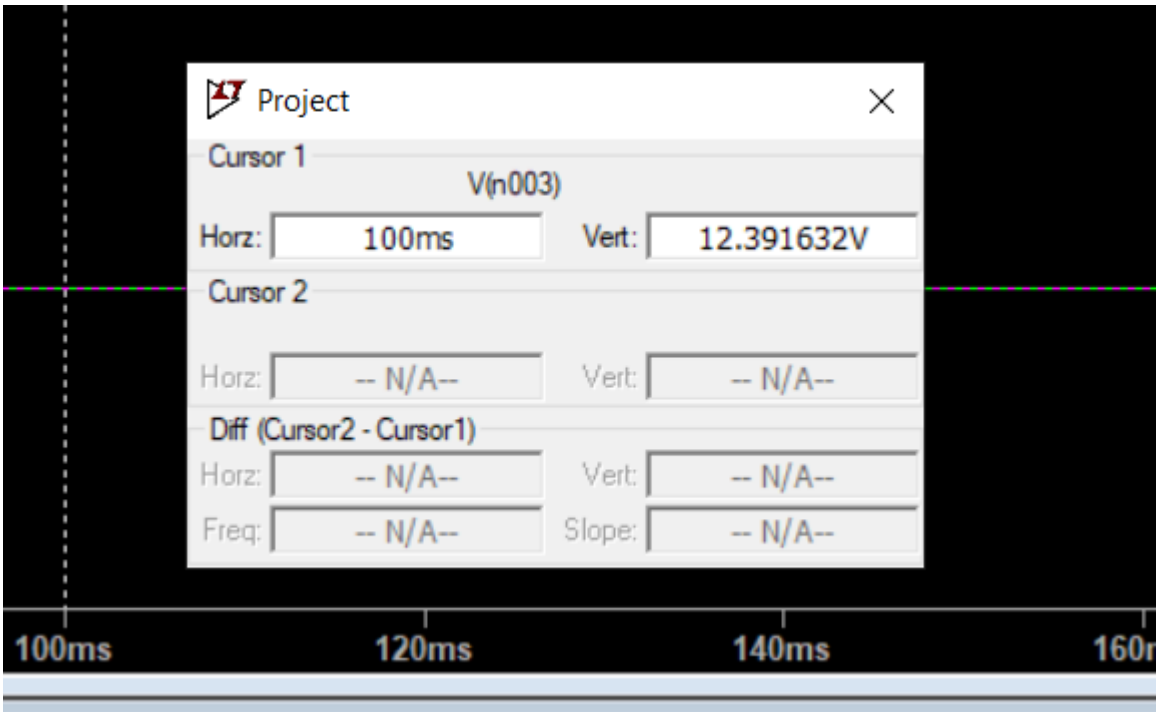


Figure 15 Output Voltage

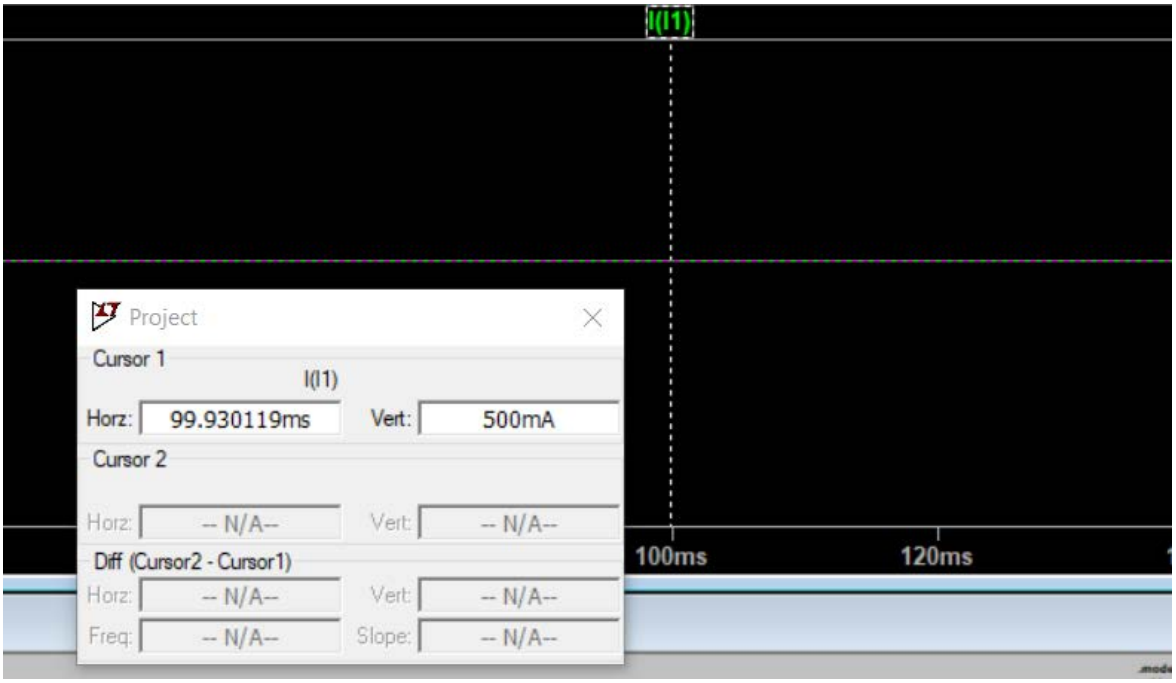


Figure 16 Output current

For the efficiency I got to the following result:

$$Vin := 15.48$$

$$Vreg := 13.69$$

$$Iin_lin := \frac{Pin_lin}{Vreg} = 0.6261$$

$$efflinF := \frac{Pout_lin}{Iin_lin \cdot Vreg} = 0.7$$

Figure 17 Linear regulator efficiency calculus

To calculate if we need a radiator for our tranzitor BD135 we have the followimg specifications:

$$P_d = 1.89W$$

$$T_{J_{max}} = 150℃$$

$$T_a = 25℃$$

$$R_{ja} = 100\ {}^{℃}/_{W}$$

$$Ta := 25$$

$$Pd := 1.89$$

$$Rja := 100$$

$$Tjmax := 150$$

$$Tamax := 25$$

$$Rjc := 10$$

$$Rpaste := 1.5$$

$$Rth_radiator := \left(\frac{Tjmax - Tamax}{Pd}\right) - Rjc - Rpaste = 54.6376$$

Figure 18 Calculations for potential radiator

Without radiator we have Rth_radiator=54.63°C/W

5.2. The integrated circuit

An important development in electronics, the integrated circuit regulators improve voltage regulation's effectiveness, dependability, and performance. Using integrated circuit (IC) technology, these small electronic devices regulate and stabilize electrical output. On a single chip, they integrate crucial parts such as a voltage divider, series regulator, error amplifier, reference voltage generator, and protective circuits.

The LM7805 is the stabilizer used in this case since the project specifications require the use of a linear stabilizer with integrated circuits that can generate a steady output voltage of 5V and a current of 0.5A.

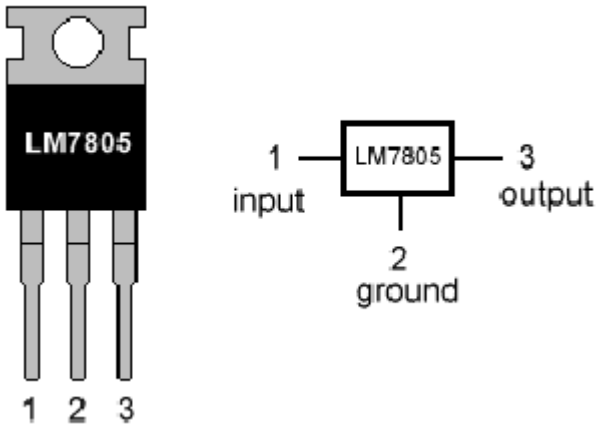


Figure 19 LM7805

For the load resistance, knowing that we needa voltage of 5V at the oputput and a 0.5A current, the calculated value for the resistance is:

$$R_S = \frac{U_{out_regulator_2}}{I_{out_regulator_2}} = \frac{5V}{0.5A} = 10\Omega$$

5.2.1. Simulations and calculations:

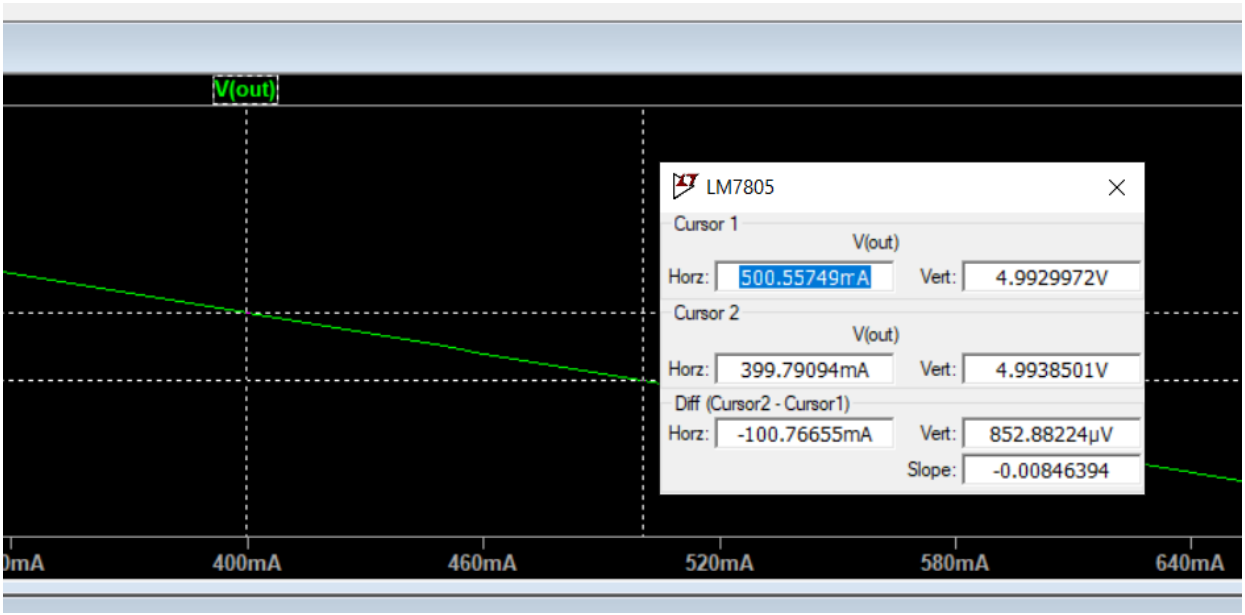


Figure 20 R0 Measured

For the measurement of Sui, I did the calculations with the result from the simulation:

$$slope_ic := 9.65 \cdot 10^{-5}$$

$$Sui := \frac{1}{slope_ic} = 10362.6943$$

Figure 21 The Sui

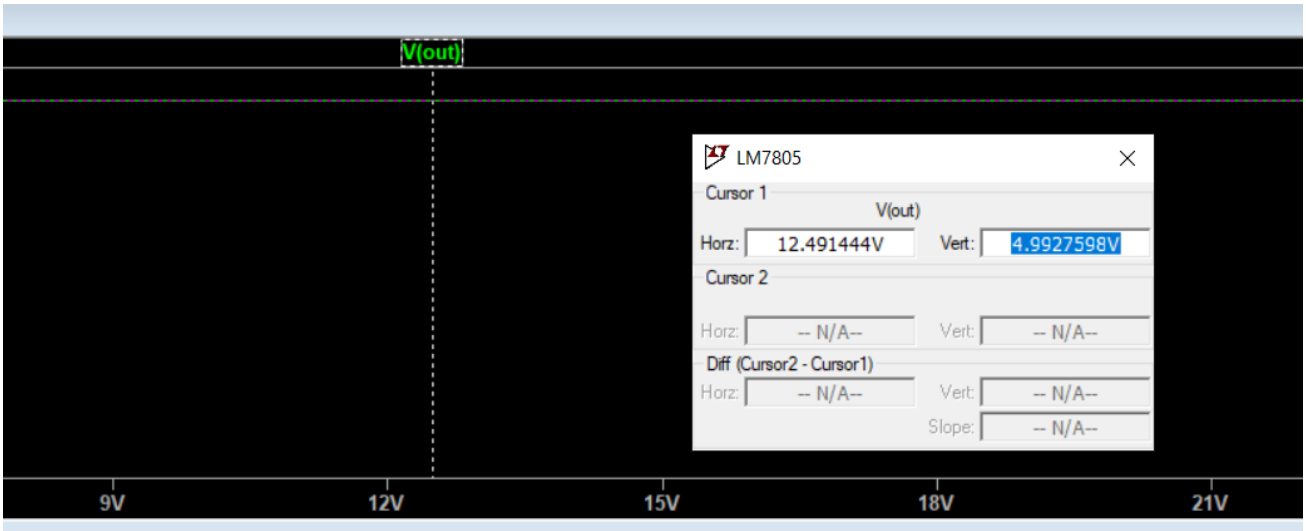


Figure 22 Output Voltage

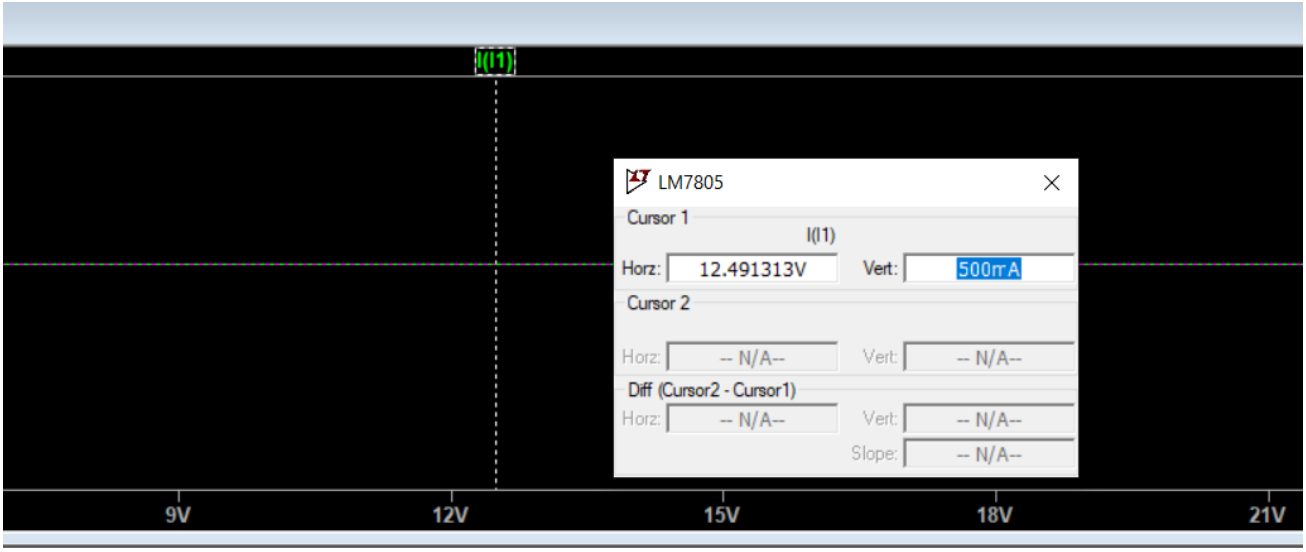


Figure 23 Output Current

To check if I need a radiator I did the calculations:

$$Pin := Us \cdot Iin = 8$$
$$Pout := 2.5\ W$$
$$Pdis := Pin - Pout = 5.5$$

$$Ta_ic := 25$$
$$Rja_ic := 19$$
$$Tj := Ta_ic + Rja_ic \cdot Pdis = 129.5$$

Figure 24 Radiator calculus

I need a heat sink with thermal resistance of 15.66°C/W or lower to safely operate the LM7805 under these conditions. Proper airflow and thermal paste will further improve performance. I choose the **WA-T220-101E** that can be bought from digikey.ro because it is a simple AUTO-CLIP Heat Sink for the TO-220 package of the IC.

5.3. The buck converter

The Buck converter, a step-down DC-DC converter made up of a transistor with a switching function, diode D, inductor L, and capacitor C, produces a voltage at its output that is lower than the input voltage. The switch in the illustration below is represented by the transistor.

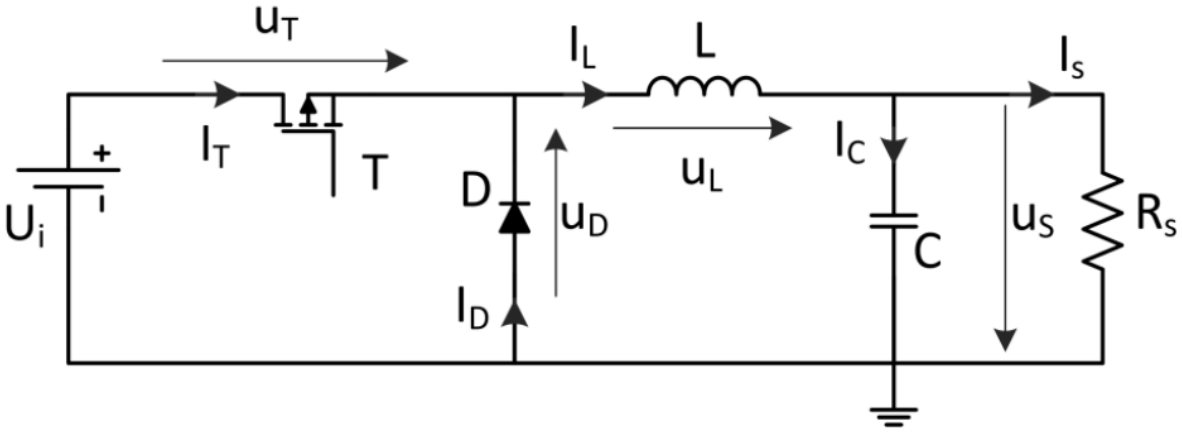


Figure 25 Schematic of the Buck converter

The output filter receives the input voltage UI when the transistor is in conduction, or the switch is on. The load current is carried by the current flowing through the inductor, which also charges the capacitor. Due to the coil's self-induced voltage, diode D opens when the transistor shuts off, giving the coil's current a channel.

Design specifications:

- Uout_regulator_3 = 3.3V
- Iout_regulator_3 = 2A
- Switching frequency = 100kHz
- Output ripple = 50mV

Sizing the components

To determine the load resistance value I used the following formula:

$$R_S = \frac{U_{out_regulator_3}}{I_{out_regulator_3}} = \frac{3.3V}{2A} = 1.65\Omega$$

To determine the duty cycle value we have the 0.5V ripple at the output of the rectifier, therefore I used the formulas:

$$\begin{cases} \delta_{min} = \dfrac{U_{out_regulator_3}}{U_{in_\text{regulator_3_max}}} = \dfrac{3.3V}{16.5V} = 0.2 \\ \delta_{max} = \dfrac{U_{out_regulator_3}}{U_{in_regulator_3_min}} = \dfrac{3.3V}{15.5V} = 0.21 \end{cases}$$

To size the inductor for the circuit to enter DCM mode, we also require the value of ISL, which is the lowest value of IL. ISL=100mA was my choice for our application. In the worst

scenario, when the input voltage is at its highest, the inductor value was selected.

For the rest I used Smath to determine the values:

$$Uout_reg3 := 3.3$$

$$Iout_min := 0.5$$

$$Uin_reg3_max := 16.5$$

$$fsw := 100 \cdot 10^{3}$$

$$Tsw := \frac{1}{fsw} = 1 \cdot 10^{-5}$$

$$Uin := 16$$

$$Vripple := 5 \cdot 10^{-2}$$

$$Lnorm := 30.70 \cdot 10^{-6}$$

$$Vin_max := 16.5$$

$$Iout_max := 3$$

$$L := \frac{Uout_reg3 \cdot Tsw}{2 \cdot Iout_min} \cdot \left(1 - \frac{Uout_reg3}{Uin_reg3_max}\right) = 2.64 \cdot 10^{-5}$$

$$\Delta IL := \frac{Uout_reg3 \cdot Tsw}{(2 \cdot L)} \cdot \left(1 - \frac{Uout_reg3}{Uin}\right) = 0.4961$$

$$Cmin := \frac{1}{8} \cdot \frac{\left(1 - \frac{Uout_reg3}{Uin}\right) \cdot Uin \cdot Tsw^{2}}{Lnorm \cdot Vripple} = 0.0001$$

$$ESR_max := \frac{(Vripple \cdot 0.75)}{\Delta IL} = 0.0756$$

And in the circuit I used a 30.70uH inductor with a 1000uF capacitor.

5.3.1. The simulations

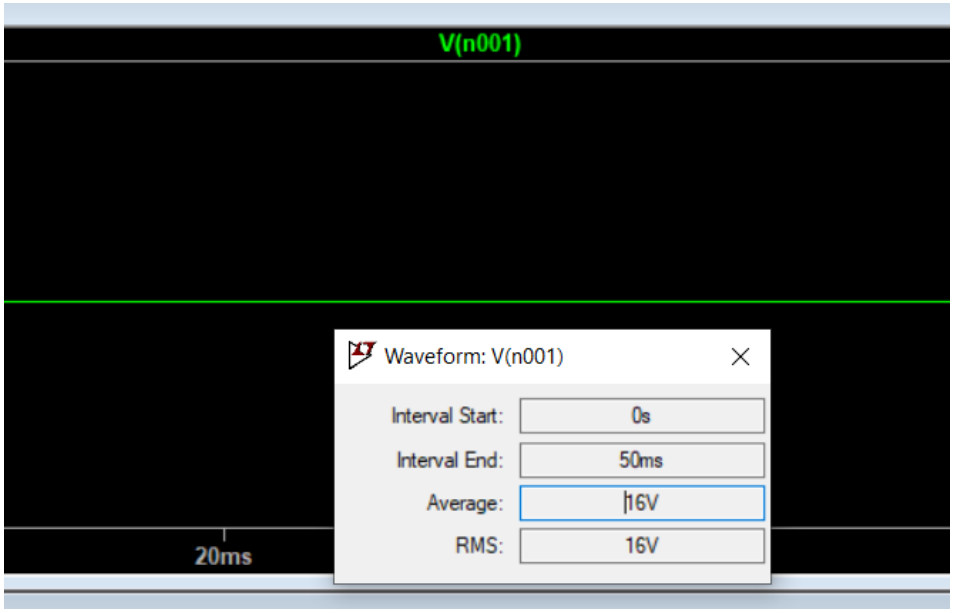


Figure 26 Input voltage

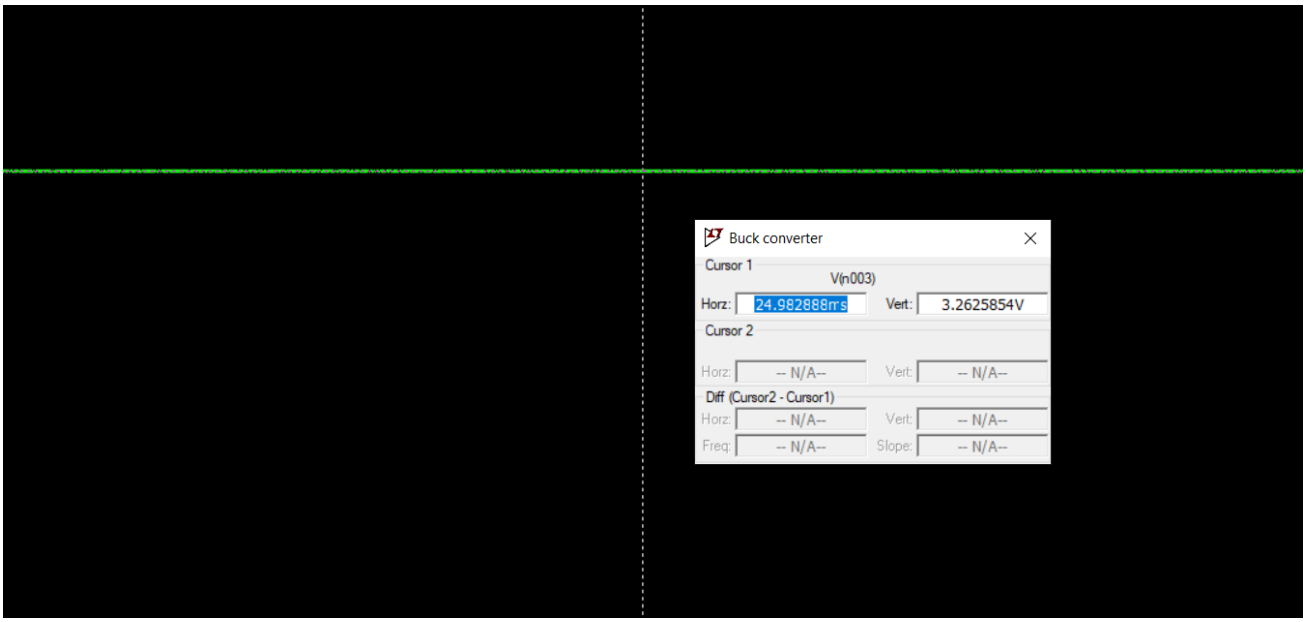


Figure 27 Output voltage

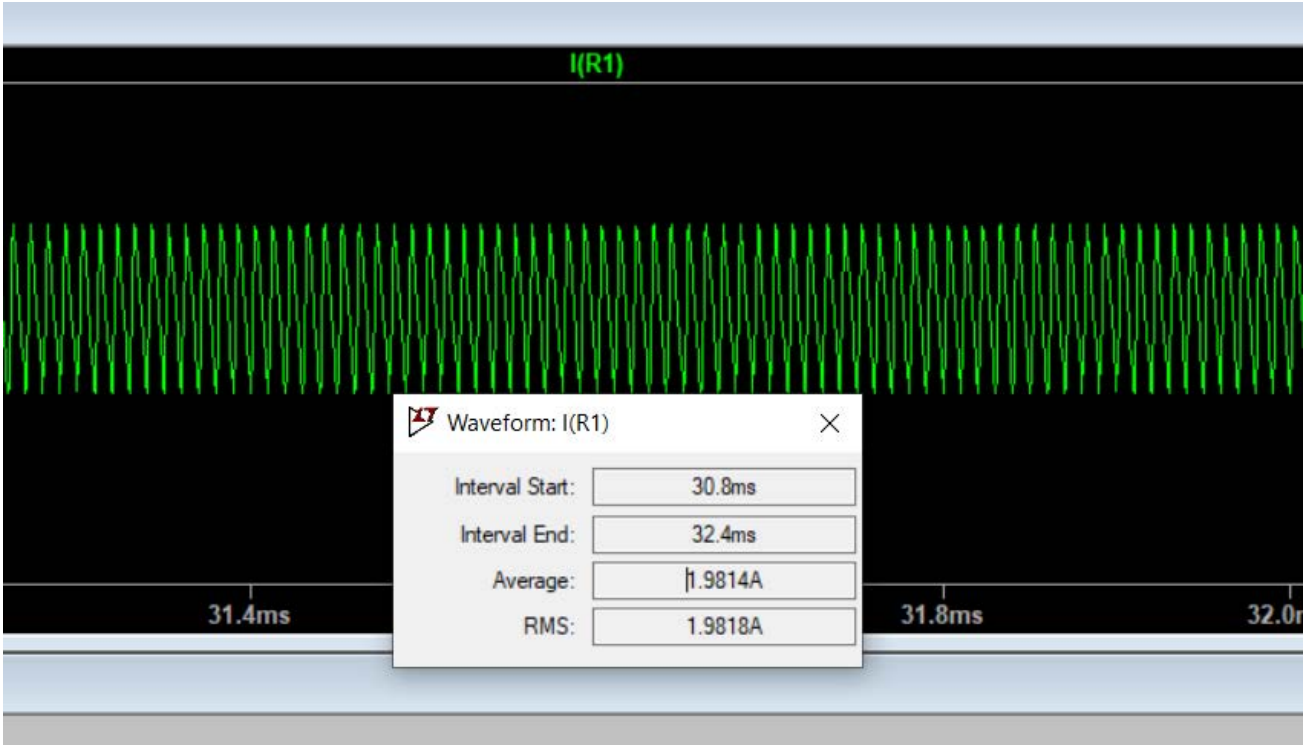


Figure 28 Output current

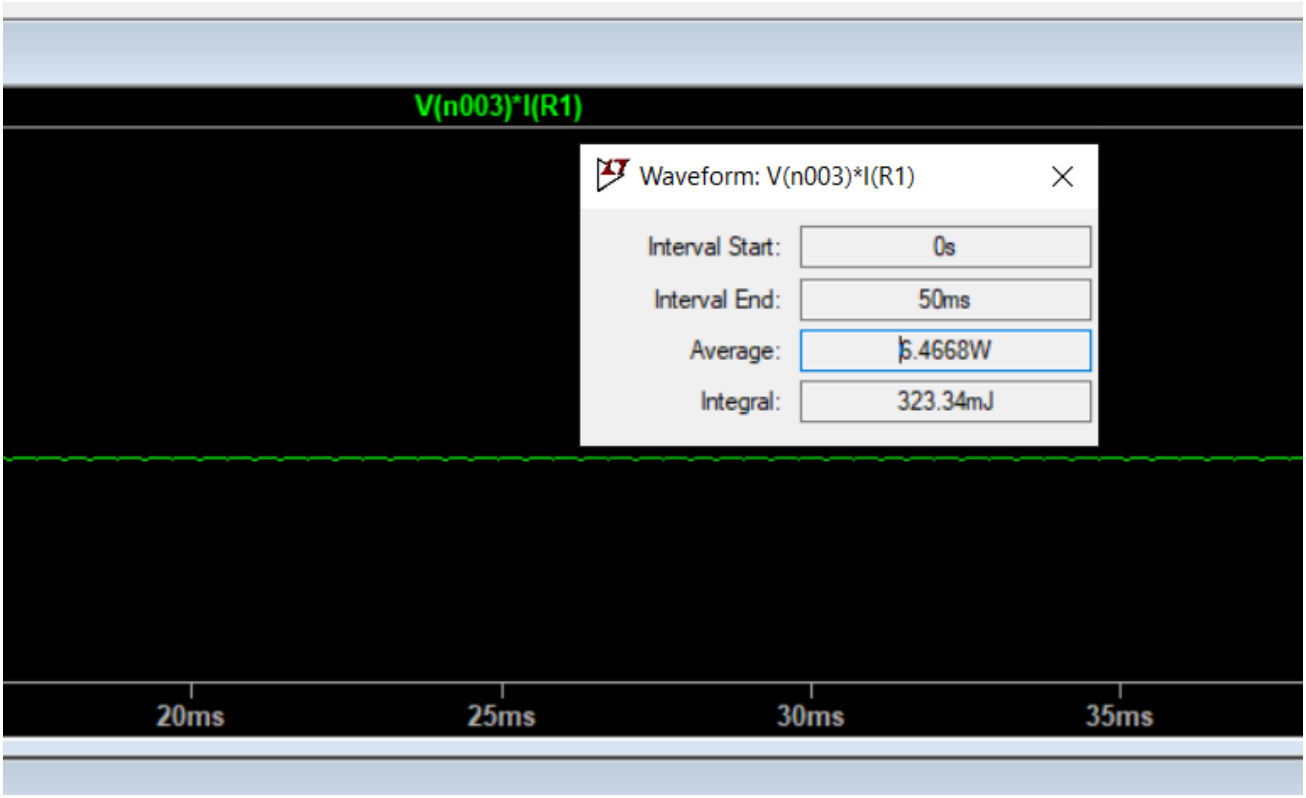


Figure 29 Output power

5.3.2. The LTSpice circuit

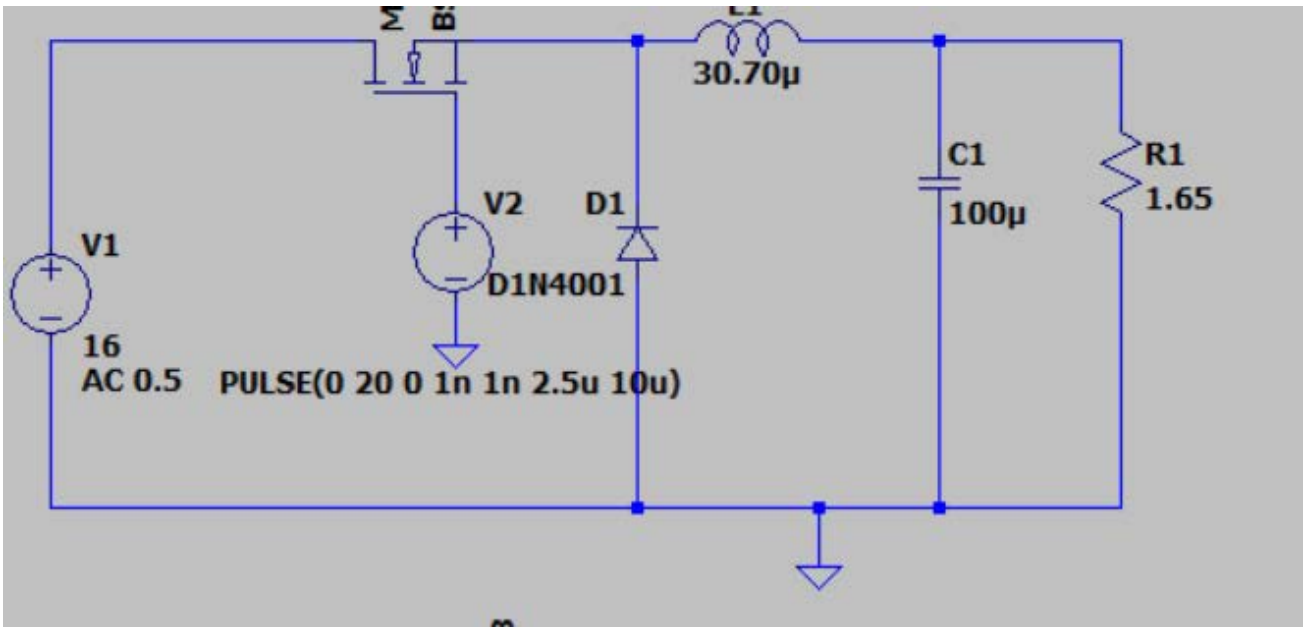


Figure 30 LTSpice implementation

5.4. Control loop

The scope of the control loop is to maintain the output despite the perturbations provoked by the load and the input voltage. To design the control loop I made the following calculations:

$$R01H := \frac{Uout_reg3}{Iout_max} = 1.1$$

$$R01L := \frac{Uout_reg3}{Iout_min} = 6.6$$

$$f_ph := \frac{1}{2 \cdot \pi \cdot \sqrt{L \cdot Cmin}} = 3045.8989$$

$$f_z := \frac{1}{2 \cdot \pi \cdot ESR_max \cdot Cmin} = 20358.5698$$

$$Kpwr := \frac{Vin_max}{Uout_reg3} = 5$$

$$Kpwrdbh := 20 \cdot \log_{10}(Kpwr) = 13.9794$$

$$fc := \frac{fsw}{10} = 10000$$

$$Q := \frac{R01H}{2 \cdot \pi \cdot f_ph \cdot L} = 2.1772$$

$$H_pwrh(\omega) := Kpwr \cdot \frac{1 + \frac{i \cdot \omega}{2 \cdot \pi \cdot f_z}}{1 + \frac{i \cdot \omega}{2 \cdot \pi \cdot f_ph} \cdot \frac{1}{Q} + \left(\frac{i \cdot \omega}{2 \cdot \pi \cdot f_ph}\right)^2}$$

$$H_pwrhfc := \left|H_pwrh(2 \cdot \pi \cdot fc)\right| = 0.563$$

$$\Phi_pwrfc := \frac{180}{\pi} \cdot \arg\left(H_pwrh(2 \cdot \pi \cdot fc)\right) = -145.0736$$

$$H_pwrhfcDB := 20 \cdot \log10\left(\left|H_pwrh(2 \cdot \pi \cdot fc)\right|\right) = -4.9897$$

$$R1 := 10\ kohm$$

$$M := 45$$

$$Boost := M - \Phi_pwrfc - 90 = 100.0736$$

$$HR := \frac{1}{H_pwrhfc} = 1.7762$$

$$Kv := \left(\tan\left(\left(\frac{Boost \cdot \pi}{4 \cdot 180}\right) + \frac{45 \cdot \pi}{180}\right)\right)^2 = 7.5637$$

$$C2 := \frac{1}{2 \cdot \pi \cdot fc \cdot HR \cdot R1} = 8.9606 \cdot 10^{-10}\ s$$

$$C1 := C2 \cdot (Kv - 1) = 5.8815 \cdot 10^{-9}\ s$$

$$R2 := \frac{\sqrt{Kv}}{2 \cdot \pi \cdot fc \cdot C1} = 7442.1938\ \Omega$$

$$R3 := \frac{R1}{Kv - 1} = 1523.5218\ \Omega$$

$$C3 := \frac{1}{2 \cdot \pi \cdot fc \cdot \sqrt{Kv} \cdot R3} = 3.7984 \cdot 10^{-9}\ s$$

After making all these calculations the ripple was still big so I made some changes at the capacitors of the control loop to fix the output ripple, therefore I got an average output voltage of 3.32V, with the same current of 2A.

5.4.1. Simulations and circuit

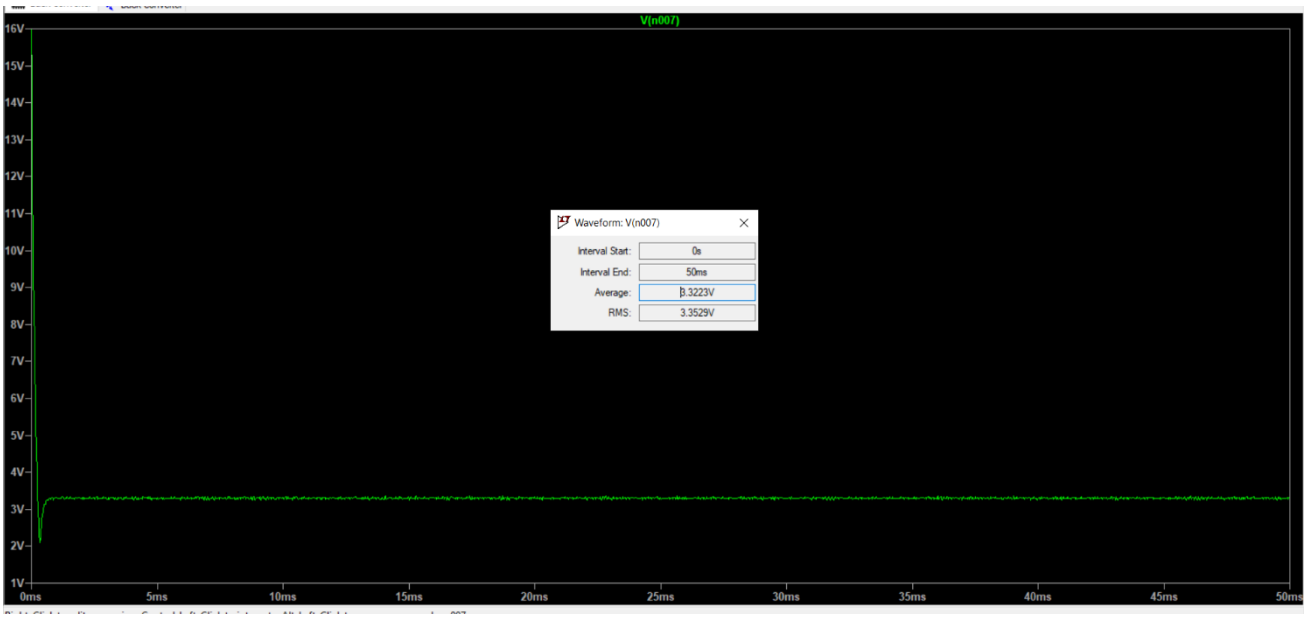


Figure 31 Output Voltage

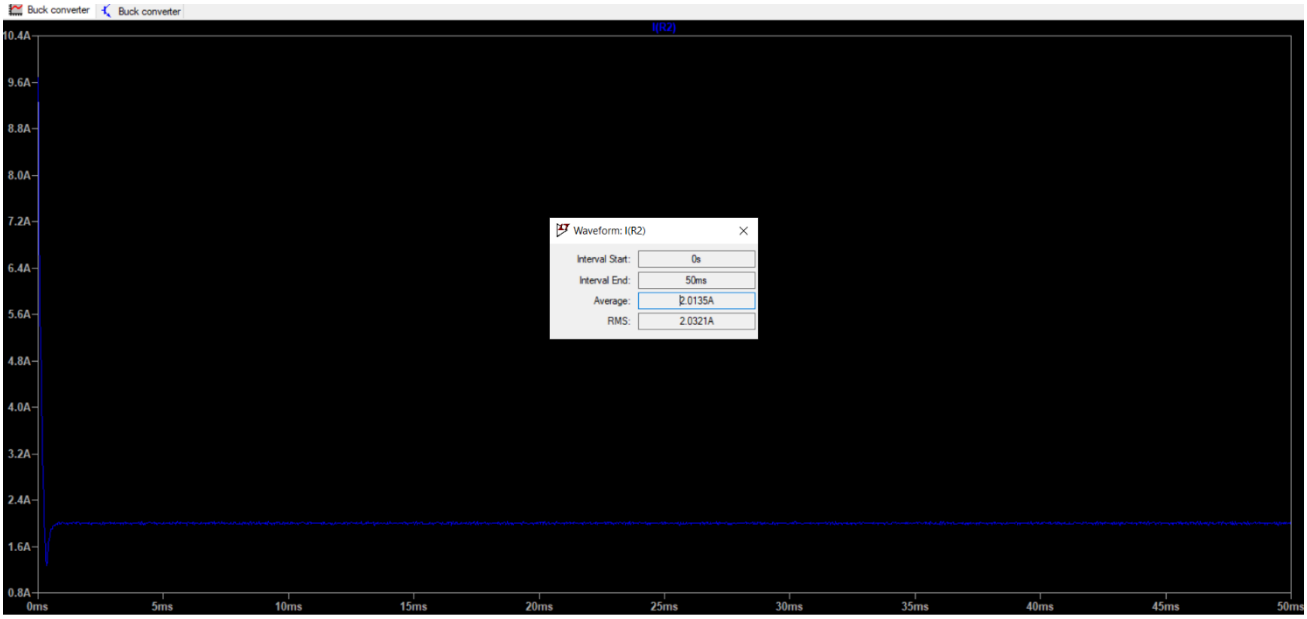


Figure 33 Output Current

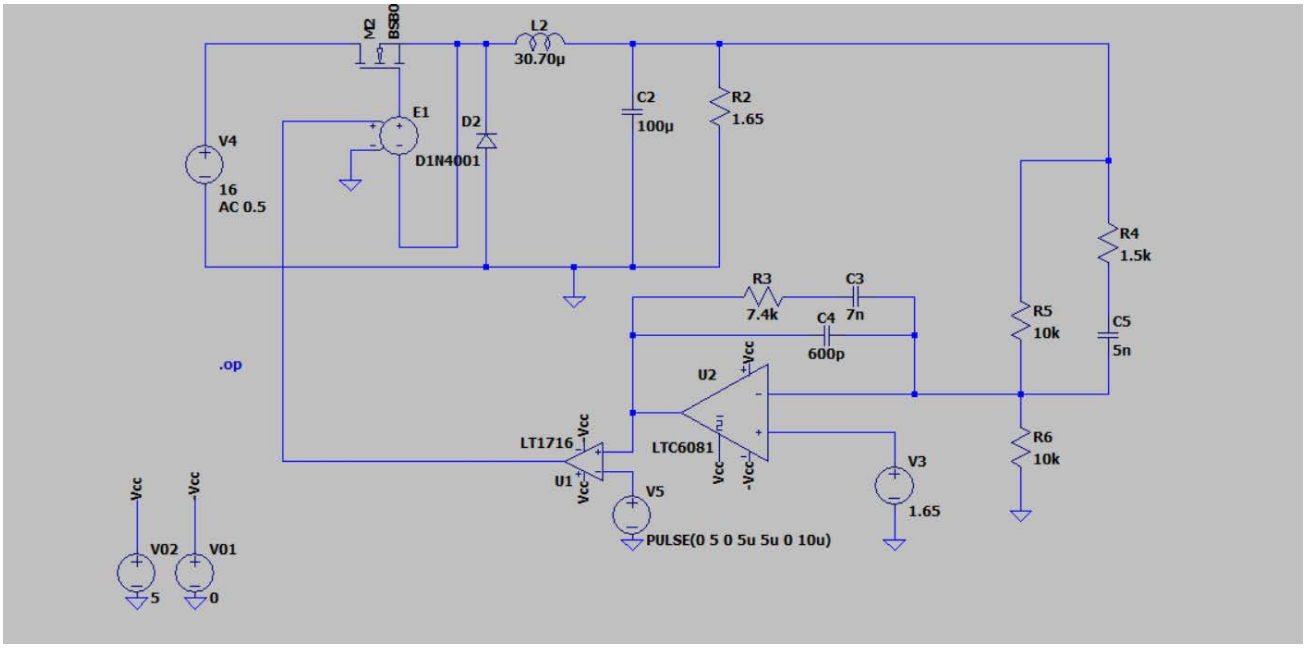


Figure 34 Buck circuit with control loop

5.5 The final circuit

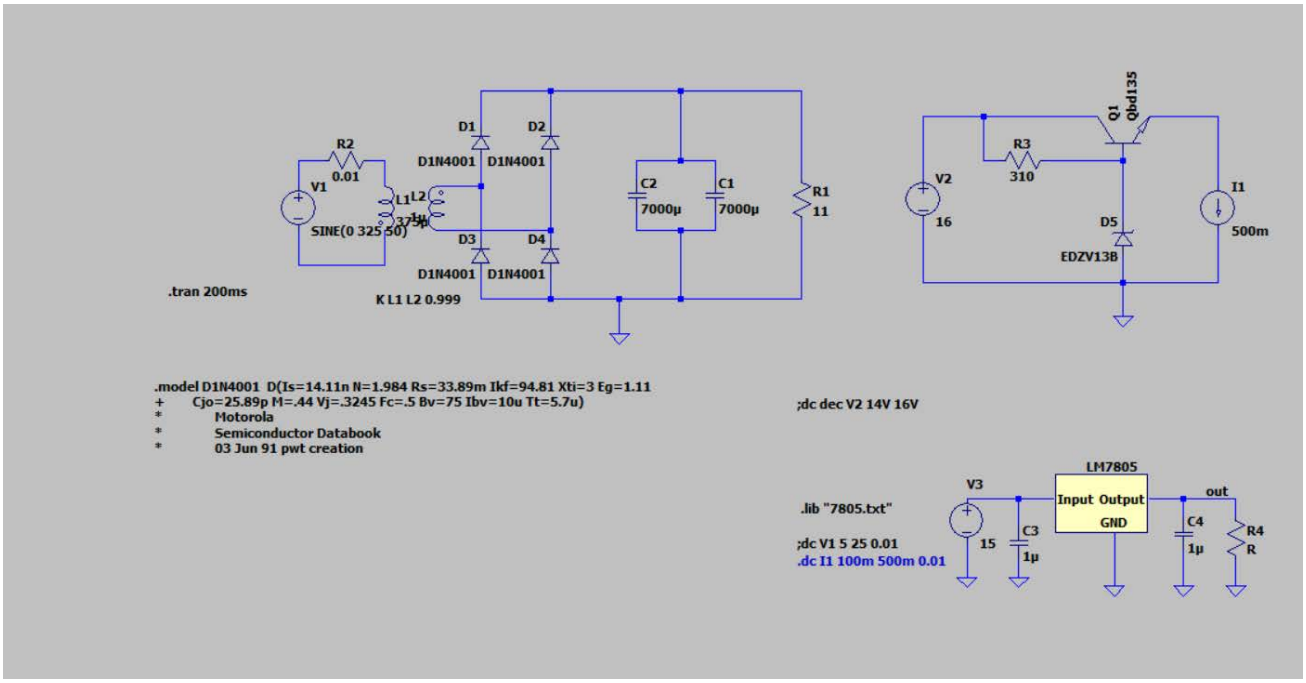


Figure 35 The first 3 parts

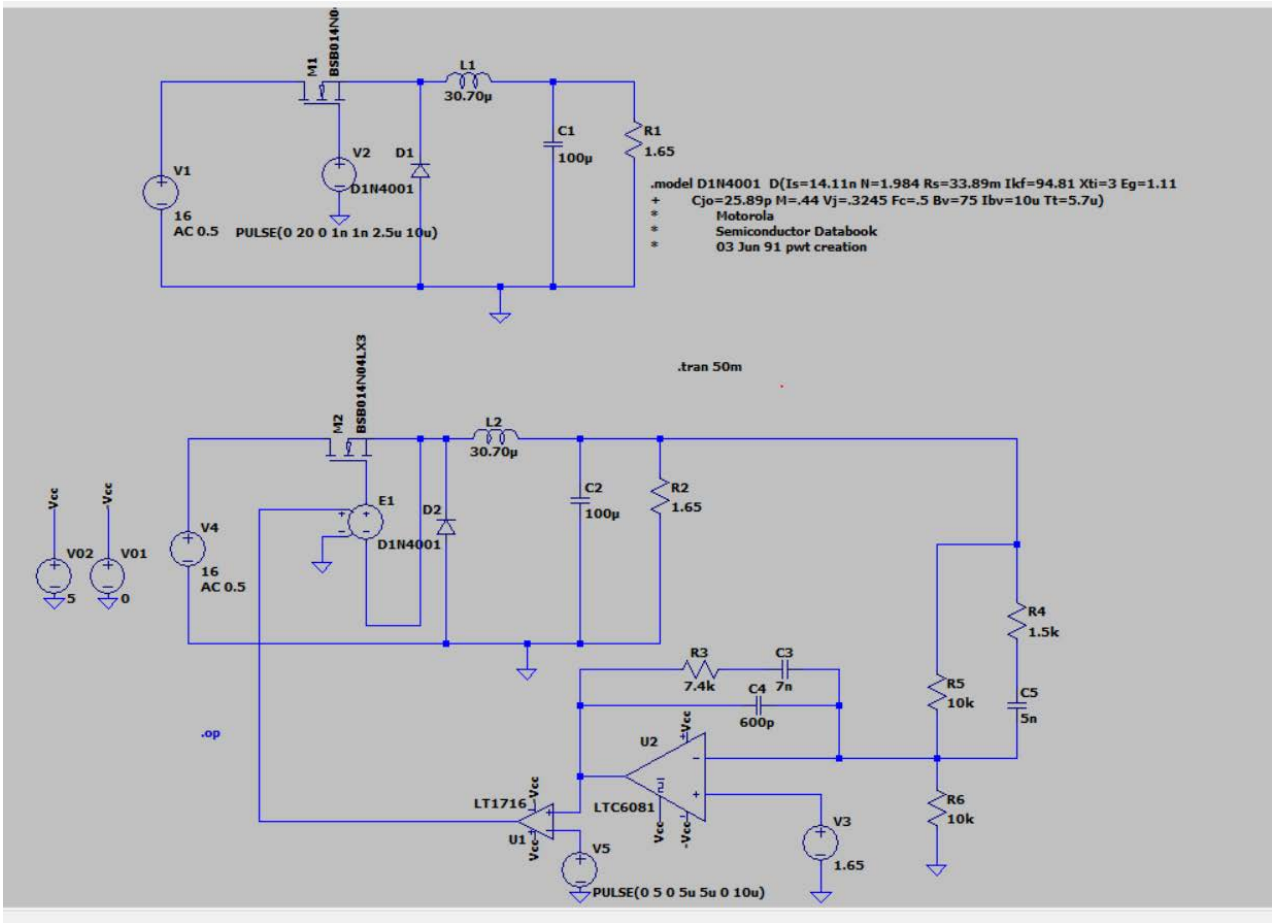


Figure 36 The last 2 parts

6. References

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